Akoizo Original Practice - Form 2 - Answer key

Division B | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) If cushion area doubles and weight stays the same, the required pressure:

A. Stays the same

B. Quadruples

C. Halves

D. Doubles

Correct: C

2. (2 points) What is the effect of missing impound?

Unit: Impound, inspection, and competition day.
Worked-case data: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

A. DS, TS, and MS are multiplied by 0.7 for each run

B. Participation points only

C. One run instead of two

D. A 10-point deduction

Correct: A

3. (2 points) What happens if one loose nickel falls off during a run?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. One construction penalty, and that nickel does not count

B. No effect

C. The run is incomplete

D. A penalty for each item lost

Correct: A

4. (2 points) What average speed is needed for a run time of 18.5 s?

A. 10 cm/s

B. 18.5 cm/s

C. 1 cm/s

D. 3.4 cm/s

Correct: A

5. (4 points) A run is incomplete with the dowel 74 cm from the finish. Find the Distance Score.

answer: DS = 40 x (185 - 74) ÷ 185 = 40 x 0.6 = 24. (4 points).

6. (4 points) Explain the difference between precision and accuracy for run times.

answer: Precision is how tightly repeated times cluster; accuracy is how close they are to the Target Time. A vehicle can be precise but consistently off target. (4 points).

7. (4 points) How does battery sag affect timing, and how can a team control for it?

answer: Lower voltage slows the motor and lengthens run time; teams measure voltage before runs, use fresh or freshly charged packs, and calibrate at that voltage. (4 points).

8. (4 points) Give one advantage and one disadvantage of a separate lift fan.

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

answer: Advantage: lift and thrust can be tuned independently. Disadvantage: the extra motor, fan, and batteries add mass. (4 points).

9. (4 points) Why is touching the vehicle mid-run so costly?

Unit: Impound, inspection, and competition day.
Worked-case data: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

answer: It sets DS, TS, and MS to zero for that run, which still counts as one of the team's runs. (4 points).

10. (4 points) Why do loose nickels need walls around them?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

answer: They slide during acceleration, bumps, and the stop, and any that fall off cost a construction penalty. (4 points).

11. (6 points) Solve this case and explain your reasoning: A hovercraft must cross 185 cm in 12.0 s at steady speed. It enters the timed zone already at speed. What speed is required, and how does adding 0.2 kg affect it?

reasoning: Average speed = 185 cm ÷ 12.0 s about 15.4 cm/s.
Extra mass lowers acceleration from the same thrust (a = F/m).
At steady speed drag balances thrust, but the vehicle reaches that speed more slowly and skirt friction may rise. (3 points).

conclusion: Recalibrate thrust whenever you change the load; the same setting gives a slower run with more nickels. (3 points).

12. (2 points) Which plan gives the highest expected score?

Supplied synthetic observations:
Run plan A: No nickels, typical error 0.3 s at TT = 12 s, always completes.
Run plan B: All nickels, typical error 0.6 s, completes 9 times in 10.
Run plan C: All nickels with ramp, error 0.2 s.

A. Plan B

B. Plan A

C. Plan C

Correct: A

13. (2 points) What setting should the team use for Run 2?

Supplied synthetic observations:
Calibration: Settings 2, 4, 6 gave 8.0, 12.0, 16.0 s.
Target Time: 13.0 s announced.
Run 1 at 4.5: 13.5 s.

A. About 5.0

B. Exactly 6

C. About 4.25

Correct: C

14. (2 points) Define “Series” in the context of Hovercraft.

answer: Cells connected end to end; voltages add. (2 points).

15. (2 points) Define “Footprint” in the context of Hovercraft.

answer: The area of the air cushion under the vehicle. (2 points).

16. (2 points) Define “Specification check” in the context of Hovercraft.

answer: Supervisor’s inspection of size, parts, and safety. (2 points).

17. (2 points) Define “Loose nickel” in the context of Hovercraft.

answer: About 5 g; each adds 0.1 to the mass term. (2 points).

18. (2 points) Define “Terminal speed” in the context of Hovercraft.

answer: Speed at which drag equals thrust and acceleration stops. (2 points).